AI Homework 1: Search Algorithms for 8-Puzzle

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1 Introduction

This report implements search algorithms to solve the 8-puzzle problem. Based on the student ID (225040014, even), the Breadth-First Search (BFS) and A^* algorithms were selected and implemented for a 1-person team.

2 Fixed Problem (Task A1 & B1)

Start State: (2, 8, 3, 1, 6, 4, 7, 0, 5)
Goal State: (1, 2, 3, 8, 0, 4, 7, 6, 5)

- **BFS Results (A1):** The path was found in 5 moves.
- **A* Results (B1):** The path was found in 5 moves using the Manhattan distance heuristic.

3 Parity Test (Task C1)

By exchanging the first two numbers (2 and 8) in the start state, the parity of the inversions count changes. The original start state inversions count was 7 (odd), while the new state (8, 2, 3...) has 6 (even). Since the goal state has 7 inversions, the states have different parity and the goal becomes unreachable.

4 Random Problem (Task A2 & B2)

Using the student ID as a random seed (225040014), the following states were generated:
Random Start: (3, 6, 0, 8, 2, 7, 4, 1, 5)
Random Goal: (5, 7, 0, 1, 2, 3, 6, 4, 8)

- **BFS Moves (A2):** 28 moves.
- **A* Moves (B2):** 28 moves.

5 Conclusion

The BFS algorithm successfully finds the shortest path but explores many nodes. The A^* algorithm is significantly more efficient when the state space grows.